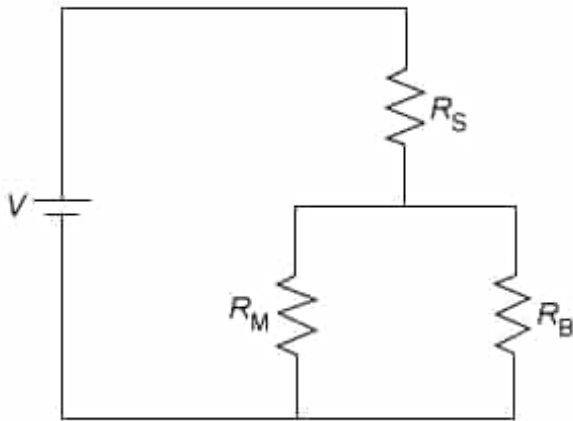


A circuit model for an electrode placed on the skin is shown.



The resistors R_S , R_M , and R_B represent the electrical properties of the skin, muscle, and blood, respectively. What is the equivalent resistance of the circuit when $V = 15$ V, $R_S = 130\ \Omega$, $R_M = 100\ \Omega$, and $R_B = 25\ \Omega$?

- ☐ A. $20\ \Omega$ [2%]
- ☐ B. $64\ \Omega$ [5%]
- ☒ C. $150\ \Omega$ [85%]
- ☐ D. $255\ \Omega$ [7%]

Correct

85% answered correctly

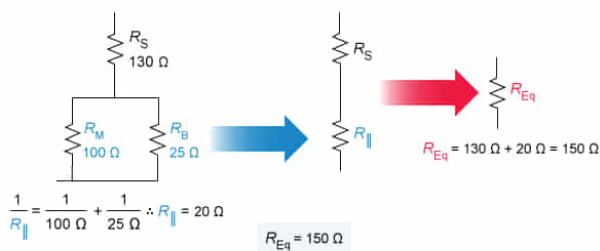
Explanation:

Calculating equivalent resistance of a circuit

$$\frac{1}{R_{\parallel}} = \frac{1}{R_M} + \frac{1}{R_B}$$

$$R_{\text{Eq}} = R_S + R_{\parallel}$$

- $R_{\parallel}$ Equivalent parallel resistance
- R_M Resistance of muscle
- R_B Resistance of blood
- R_{Eq} Equivalent resistance
- R_S Resistance of skin



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A circuit consisting of a network of **resistors** can typically be simplified by combining resistors connected in **series** and resistors connected in **parallel** into **equivalent resistors** R_{Eq} . The R_{Eq} for resistors connected in series equals the sum of the individual resistors:

$$R_{\text{Eq}} = R_1 + R_2 + R_3 + \dots$$

On the other hand, the inverse of the R_{Eq} for resistors connected in parallel equals the sum of the inverses of the individual resistors:

$$\frac{1}{R_{\text{Eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

In this question, R_{M} and R_{B} are connected in parallel and their R_{Eq} is connected in series with R_{S} . Combining R_{M} and R_{B} yields a resistor value of $R_{\parallel}$ given by:

$$\frac{1}{R_{\parallel}} = \frac{1}{R_{\text{M}}} + \frac{1}{R_{\text{B}}} = \frac{1}{100 \, \Omega} + \frac{1}{25 \, \Omega}$$

$$\frac{1}{R_{\parallel}} = \frac{1}{100 \, \Omega} + \frac{4}{100 \, \Omega} = \frac{5}{100 \, \Omega}$$

$$R_{\parallel} = \frac{100 \, \Omega}{5} = 20 \, \Omega$$

Furthermore, combining $R_{\parallel}$ and R_{S} in series yields R_{Eq} of:

$$R_{\text{Eq}} = R_{\text{S}} + R_{\parallel} = 130 \, \Omega + 20 \, \Omega = 150 \, \Omega$$

Therefore, R_{Eq} for the combination of R_{S} , R_{M} , and R_{B} equals 150 Ω .

(Choice A) 20 Ω is the equivalent resistance of R_{M} and R_{B} instead of the equivalent resistance of all the resistors.

(Choice B) 64 Ω is calculated by switching the equivalent resistance equations for series and parallel resistors.

(Choice D) 255 Ω is obtained by summing all the resistor values instead of using the equivalent resistance equations for resistors connected in series and in parallel.

Educational objective:

Resistors connected in parallel can be replaced with an equivalent resistance calculated from the equation $1/R_{\text{Eq}} = 1/R_1 + 1/R_2 + \dots + 1/R_n$.

Time spent: 0 sec

Subject:
Physics

QID: 403718

System:
Electrostatics and Circuits